सहायक प्राध्यापक (संगणकशास्त्र) महाराष्ट्र शिक्षण सेवा, गट-अ
(महाविद्यालयीन शाखा)
उच्च व तंत्र शिक्षण विभाग
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परीक्षेचे टप्पे: १) लेखी परीक्षा - २०० गुण २) मुलाखत- ५० गुण

-: परीक्षा योजना :-

विषय व सांकेतांक	माध्यम	प्रश्नसंख्या	गुण	कालावधी	दर्जा	प्रश्नपत्रिकेचे स्वरुप
विषयांशी संबंधीत घटक (११६६)	इंग्रजी	१००	२००	एक तास	पदव्युत्तर पदवी	वस्तुनिष्ठ बहुपर्यायी

अ) नकारात्मक गुणदान -

१) प्रत्येक चुकीच्या उत्तराकरीता २५% किंवा १/४ एवढे गुण एकूण गुणांमधून वजा/ कमी करण्यात येतील.
२) एखाद्या प्रश्नाची एकापेक्षा अधिक उत्तरे दिली असल्यास अथवा ज्या उमेदवाराने उत्तरपत्रिकेत पूर्ण वर्तुळ चिन्हांकित केले नसेल अशा प्रश्नाचे उत्तर चुकीचे समजण्यात येऊन त्या प्रश्नाच्या उत्तराकरीता २५% किंवा १/४ एवढे गुण एकूण गुणांमधून वजा/कमी करण्यात येतील.
३) वरीलप्रमाणे कार्यपध्दतीचा अवलंब करताना एकूण अंतिम गुणांची बेरीज अपूर्णांकात आली तरीही ती अपूर्णांकातच राहील व पुढील कार्यवाही त्याच्या आधारे करण्यात येईल.
४) एखाद्या प्रश्नाचे उत्तर अनुत्तरित असेल तर, अशा प्रकरणी नकारात्मक गुणांची पध्दत लागू असणार नाही.

ब) अंतिम गुणवत्ता यादी ही वस्तुनिष्ठ परीक्षेतील व मुलाखतीतील एकत्रित गुणांवर आधारीत राहील.

-: अभ्यासक्रम :-

विषयांशी संबंधित घटक यामध्ये खालील घटक व उपघटकांचा समावेश असेल.

अ.क्र.	घटक व उपघटक
1	**Discrete Structures** Mathematical Logic, Sets and Relations, Counting, Mathematical Induction and Discrete Probability, Group Theory, Graph Theory, Boolean Algebra.
2	**Computer System Architecture** Digital Logic Circuits and Components, Data Representation, Register Transfer and Micro operations, Basic Computer Organization and Design, Programming the Basic Computer, Microprogrammed Control, Central Processing Unit, Pipeline and Vector Processing, Input-Output Organization, Memory Hierarchy, Multiprocessors.
3	**Programming Languages** Language Design and Translation Issues, Elementary Data Types, Programming in C, Object Oriented Programming in C++.
4	**Computer Graphics** Video-Display Devices, Raster-Scan and Random-Scan Systems; Graphics Monitors, Input Devices, Points and Lines; Line Drawing Algorithms, Mid-Point Circle and Ellipse Algorithms; Scan Line Polygon Fill Algorithm, Boundary-Fill and Flood- Fill, 2-D Geometrical Transforms and Viewing, 3-D Object Representation, Geometric Transformations and Viewing.
5	**Database Management Systems** Database System Concepts and Architecture, Data Modeling, Database Design, ER Diagrams, Relational Algebra and Calculus, SQL, Normalization for Relational Databases, Enhanced Data Models, Transaction Management and Concurrency control, Data Warehousing and Data Mining, NOSQL.
6	**System Software and Operating System** System Software, Basics of Operating Systems, Process Management, Threads, CPU Scheduling, Deadlocks, Memory Management, Storage Management, File and Input/Output Systems, Security, Virtual Machines, Linux Operating Systems, Windows Operating Systems, Distributed Systems

7	**Software Engineering** Software Process Models, Software Requirements, Software Design, Software Quality, Estimation and Scheduling of Software Projects, Software Testing, Software Configuration Management
8	**Data Structures and Algorithms** Data Structures: Arrays and their Applications; Sparse Matrix, Stacks, Queues, Priority Queues, Linked Lists, Trees, Forest, Binary Tree, Threaded Binary Tree, Binary Search Tree, AVL Tree, B Tree, B+ Tree, B* Tree, Data Structure for Sets, Graphs, Sorting and Searching Algorithms; Hashing. Performance Analysis of Algorithms and Recurrences, Design Techniques, Lower Bound Theory, Graph Algorithms, Complexity Theory, Greedy method techniques, Dynamic programming techniques, Divide and Conquer.
9	**Theory of Computation and Compilers** Theory of Computation, Regular Language Models, Context Free Language, Turing Machines (TM), Unsolvable Problems and Computational Complexity, Syntax Analysis, Semantic Analysis, Run Time System, Intermediate Code Generation, Code Generation and Code Optimization.
10	**Computer Networks** Components of a Data Communication System, Simplex, Half- Duplex and Duplex Modes of Communication; Multiplexing, Transmission Media, Transmission Errors, Error Handling Mechanisms, Network Topologies, Local Area Networks, Metropolitan Area Networks, Wide Area Network, Wireless Networks, Internet. Layered Architecture, OSI Reference Model and its Protocols; TCP/IP Protocol Suite,Physical, Logical, Port and Specific Addresses; Switching Techniques. Functions of OSI and TCP/IP Layers, IPv4 Structure and Address Space, IPv6.
11	**Machine Learning and Artificial Intelligence (AI)** **Approaches to AI, Knowledge Representation, Planning:** Components of a Planning System, Linear and Non Linear Planning; Goal Stack Planning, Hierarchical Planning, STRIPS, Partial Order Planning. **Machine Learning :** Regression, Classification, Supervised and Unsupervised Learning, Support Vector Machine, K-Nearest Neighbor, Hidden Markov Model, Performance measurement metrics. **Artificial Neural Networks (ANN):** Supervised, Unsupervised and Reinforcement Learning; Single Perceptron, Multi-Layer Perceptron, Self-Organizing Maps, Hopfield Network.

दिनांक – २१ जुलै, २०२५

अवर सचिव
महाराष्ट्र लोकसेवा आयोग